

```
#include <iostream>
```

```
int main() {
```

```
    double power_factor;
```

```
    std::cout << "Enter the power factor of the electrical load: ";
```

```
    std::cin >> power_factor;
```

```
    if (power_factor < 0.0 || power_factor > 1.0) {
```

```
        std::cout << "Status: Invalid power factor" << std::endl;
```

```
    } else if (power_factor >= 0.0 && power_factor <= 0.50) {
```

```
        std::cout << "Status: Poor power factor" << std::endl;
```

```
    } else if (power_factor >= 0.51 && power_factor <= 0.80) {
```

```
        std::cout << "Status: Fair power factor" << std::endl;
```

```
    } else if (power_factor >= 0.81 && power_factor <= 0.95) {
```

```
        std::cout << "Status: Good power factor" << std::endl;
```

```
    } else {
```

```
        std::cout << "Status: Excellent power factor" << std::endl;
```

```
    }
```

```
    return 0;
```

```
}
```